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The Editor-in-Chief
Ear and Hearing
Wolters Kluwer Health

Dear Editor-in-Chief,

I am pleased to submit our manuscript, “A Fuzzy Logic Framework for Sensorineural Hearing Loss Classification: Preserving Diagnostic Gradation Lost to Crisp Thresholds”, for consideration as an original article in Ear and Hearing. The blinded manuscript, figures, and tables are uploaded as separate files as required by the submission guidelines.

Ear and Hearing has long been the home for rigorous work at the interface of clinical audiology and quantitative methods, and we believe this paper speaks directly to your readership. The way we classify hearing loss from pure-tone audiometry has changed little since the WHO thresholds were introduced decades ago, yet the crisp boundaries at 25, 40, 55, 70, and 90 dB HL force a continuous biological variable into discrete categories and misclassify the substantial proportion of patients whose thresholds fall near a boundary. Our work offers a graded alternative built on fuzzy logic, a transparent and interpretable framework well suited to clinical decision support.

We developed a Mamdani-type fuzzy inference system with a 48-rule audiology-derived rule base that maps frequency-specific thresholds into a continuous Fuzzy Audiometric Index (FAI) on a 0 to 100 scale, together with configuration typing, an asymmetry index, and a temporal tracking module. The membership functions were optimised on the training split of a combined adult cohort from three NHANES audiometry cycles (1999–2000, 2011–2012, 2015–2016; 10,889 participants aged 20–69 years) and the system was validated on a held-out test set of 3,912 ears from 1,966 participants, split at the participant level so that no participant contributes ears to both training and test. Single-ear validation omits the asymmetry-anchor rule (a

documented artifact: with one ear the asymmetry input is zero, so the anchor rule would fire at full strength for every ear and compress the FAI scale); bilateral use in the web application retains the full 48-rule base. Against the WHO PTA-4 reference it achieved a weighted Cohen's kappa of 0.93, with 98.1% agreement in clear cases. In borderline cases within ± 5 dB of a severity boundary — 18.7% of test ears — the FAI provides a graded, continuous classification; agreement with the crisp reference at these boundaries was 79.8%, reflecting the very gradation the framework is designed to preserve. Configuration typing is provided as a graded six-category membership vector, and the longitudinal module detected sub-threshold progression that stepwise grades miss. Unlike machine learning comparators, the fuzzy rule base remains fully inspectable and modifiable by clinicians.

We believe the manuscript offers three things Ear and Hearing readers will find useful. First, a demonstration that graded classification materially improves borderline case resolution by preserving gradation that crisp thresholds discard, while retaining substantial overall agreement with the reference standard. Second, a single framework that integrates severity, configuration, asymmetry, and temporal tracking, which earlier fuzzy applications in audiology have not attempted. Third, validation on a large population-based dataset with a proper train-test split, which has been lacking in this literature.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The analyses use de-identified public data from NHANES, for which no ethics approval was required. Data and analysis code are available from the corresponding author on request. I declare no conflicts of interest, and as sole author I have read and approved the submitted version. In accordance with journal policy, I disclose that a large language model-based assistant was used during manuscript preparation (drafting, editing, and code organisation); this use is described in the manuscript's Methods and Acknowledgements, and the author takes full responsibility for all content.

Thank you for considering our work. We would welcome the opportunity to revise the manuscript in response to reviewer feedback.

Yours sincerely,

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